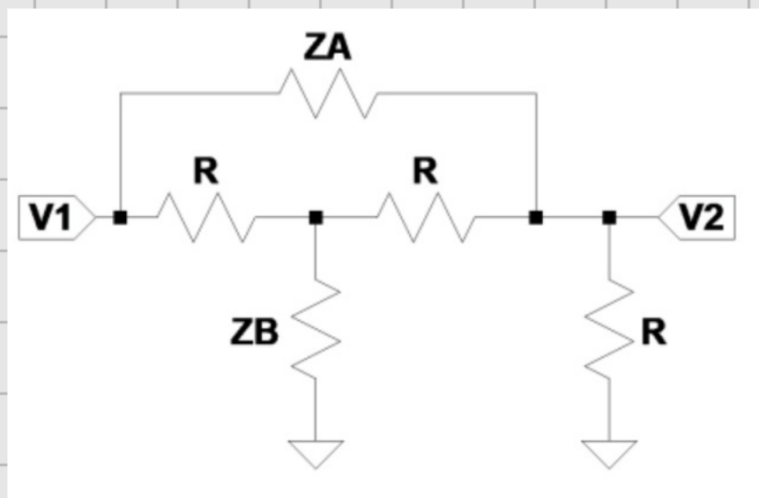
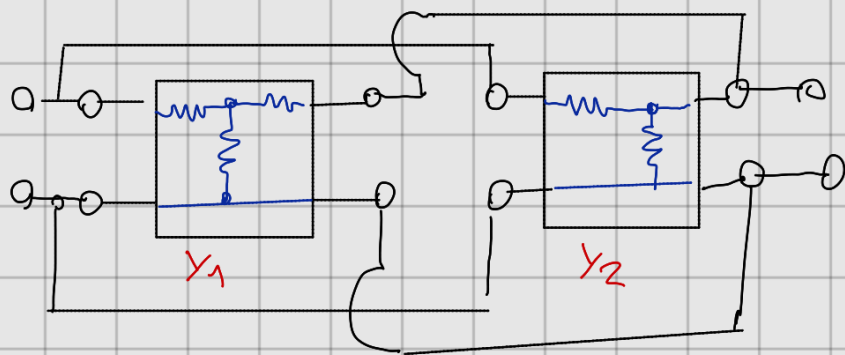
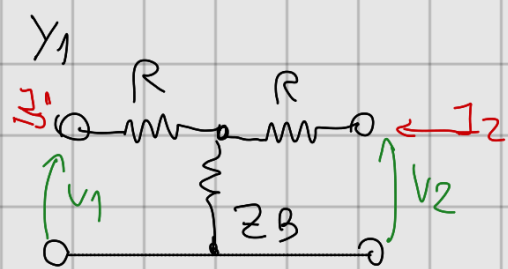




Se puede dividir en 2 cuadripolos, uno de ellos tee.



como están en paralelo, trabajo con parámetros ADMITANCIA

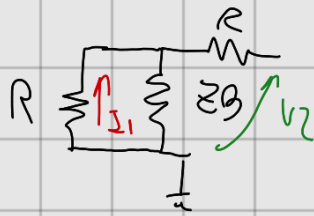


$$Y_{11} = \frac{I_1}{V_1} \Big|_{V_2=0} = (G + Y_B) \parallel G$$

$$Y_{11} = \frac{\frac{R+Z_B}{R \cdot Z_B} \cdot \frac{1}{R}}{\frac{1}{R} + \frac{R+Z_B}{R \cdot Z_B}}$$

$$Y_{11} = \frac{\frac{R+Z_B}{R \cdot Z_B}}{\frac{Z_B + R + Z_B}{R \cdot Z_B}} = \frac{R+Z_B}{(R+2Z_B) \cdot R}$$

$$Y_{12} = \frac{I_1}{V_2} \Big|_{V_1=0}$$



$$I_1 \Big|_{V_1=0} = V_2 \cdot \frac{R \parallel Z_B}{R \parallel Z_B + R} \cdot \frac{1}{R} \cdot (-1)$$

$$Y_{12} = -\frac{1}{R} \cdot \frac{\frac{R \cdot Z_B}{R + Z_B}}{\frac{R \cdot Z_B}{R + Z_B} + R} = -\frac{1}{R} \cdot \frac{R \cdot Z_B}{2R \cdot Z_B + R^2} = -\frac{Z_B}{2R \cdot Z_B + R^2}$$

$$Y_{21} = \frac{I_2}{V_1} \Big|_{V_2=0}$$



$$I_2 \Big|_{V_2=0} = V_1 \cdot \frac{R \parallel Z_B}{R \parallel Z_B + R} \cdot \frac{1}{R} \cdot (-1) \quad \text{idem } Y_{12}..$$

$$Y_{21} = Y_{12}$$

$$Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0}$$



$$Y_{22} = (R \parallel Z_B + R)^{-1}$$

$$Y_{22} = \left(\frac{R \cdot Z_B}{R + Z_B} + R \right)^{-1} = \left(\frac{R \cdot Z_B + R^2 + R \cdot Z_B}{R + Z_B} \right)^{-1} = \frac{R + Z_B}{R^2 + 2R \cdot Z_B}$$

$$Y_1 = \begin{pmatrix} \frac{R + Z_B}{R(R + 2Z_B)} & -\frac{Z_B}{R(R + 2Z_B)} \\ -\frac{Z_B}{R(R + 2Z_B)} & \frac{R + Z_B}{R(R + 2Z_B)} \end{pmatrix}$$

luego, Y_2 ..



$$Y_{21} = \frac{I_1}{V_1} \Big|_{V_2=0} = \frac{1}{Z_A}$$

$$Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0} = -\frac{1}{Z_A}$$



$$Y_{21} = \frac{I_2}{V_1} \Big|_{V_2=0} = -\frac{1}{Z_A}$$



$$Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0} = \frac{1}{Z_A} + \frac{1}{R} = \frac{Z_A + R}{Z_A \cdot R}$$

Entonces

$$Y_2 = \begin{bmatrix} 1/Z_A & -1/Z_A \\ -1/Z_A & (Z_A + R)/Z_A \cdot R \end{bmatrix}$$

FINALMENTE, Y_{TOTAL} ..

$$Y_{TOT} = Y_1 \cdot Y_2$$

$$Y_{TOT} = \frac{1}{R(R+Z_B)} \cdot \begin{bmatrix} R+Z_B & -Z_B \\ -Z_B & R+Z_B \end{bmatrix} \cdot \frac{1}{Z_A} \begin{bmatrix} 1 & -1 \\ -1 & \frac{Z_A+R}{R} \end{bmatrix}$$

$$Y_{TOT} = \frac{1}{R(R+2Z_B) \cdot Z_A} \begin{pmatrix} R+2Z_B & -R-Z_B - \frac{Z_B(Z_A+R)}{R} \\ -R-2Z_B & Z_B + \frac{(R+Z_B)(Z_A+R)}{R} \end{pmatrix}$$

$$Y_{TOT} = \begin{pmatrix} \frac{1}{R \cdot Z_A} & \frac{-R - Z_B - R \cdot Z_B \cdot (Z_A+R)}{R \cdot Z_A \cdot (R+2 \cdot Z_B)} \\ -\frac{1}{R \cdot Z_A} & \frac{Z_B + (R+Z_B)(R+Z_A) \cdot R}{R \cdot Z_A \cdot (R+2Z_B)} \end{pmatrix}$$

Si $R=1, Z_A=Z_B=1$..

$$Y_{TOT} = \begin{pmatrix} 1 & -\frac{4}{3} \\ -1 & \frac{5}{3} \end{pmatrix}$$